MAIS PAC

IPhO 02.03: Dynamics: Energy, Momentum and Rotation · Lecture + Problems

Dynamics: Energy, Momentum and Rotation

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Newton's second law is about change, not motion

$$F_{\text{net}} = \frac{d p}{dt} = m a \quad (\text{constant } m). \tag{1}$$

A force does not **cause** motion; it causes **change of** motion. That single sentence disposes of most beginner errors. In components, the law is a pair of independent scalar equations, and problems become bookkeeping — provided you drew the free-body diagram first.

Three equivalent bookkeeping systems are available for any mechanics problem:

- **Force and acceleration** — best when you need the motion at every instant.
- **Energy** — best when the question only cares about speeds and heights.
- **Momentum** — best when forces are large, brief and unknown (collisions, explosions, kicks).

Strong solutions choose the cheapest system, not the most familiar one.

2. Work, energy and the area under a graph

Work done by a force along a displacement is

$$W = \int F \cdot d r, \tag{2}$$

which for a constant force is Fdcos θ, and graphically is the **area under an F–x graph**. Power is the rate of doing work, P = F· v.

For a spring, F = kx, so the stored energy is the triangular area

$$U = (1)/(2)\ kx^2. \tag{3}$$

For a uniform gravitational field near the surface, U = mgh. For the inverse-square field, U = -GMm/r — the minus sign is what makes bound orbits possible.

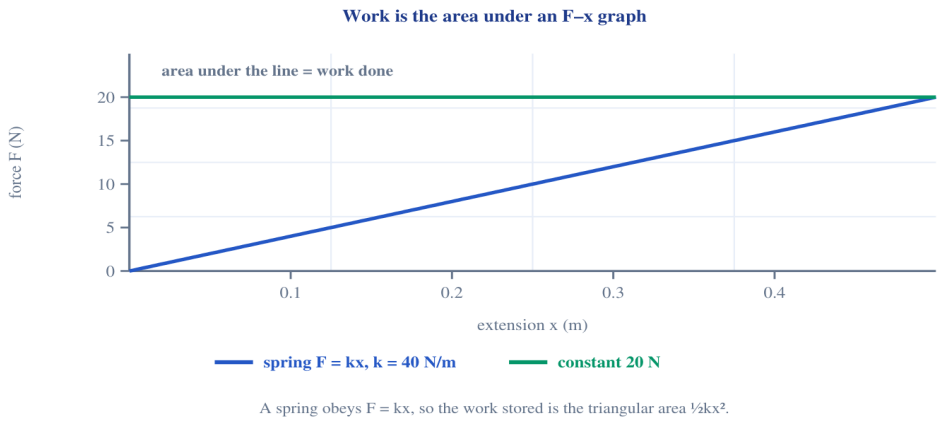


Figure. A spring obeys F = kx, so the work stored is the triangular area ½kx².

The work–energy theorem follows from Newton's law and is the workhorse of olympiad mechanics:

$$W_{\{net\}} = \Delta K. \tag{4}$$

With only conservative forces, K + U = {constant} and the problem collapses to one equation.



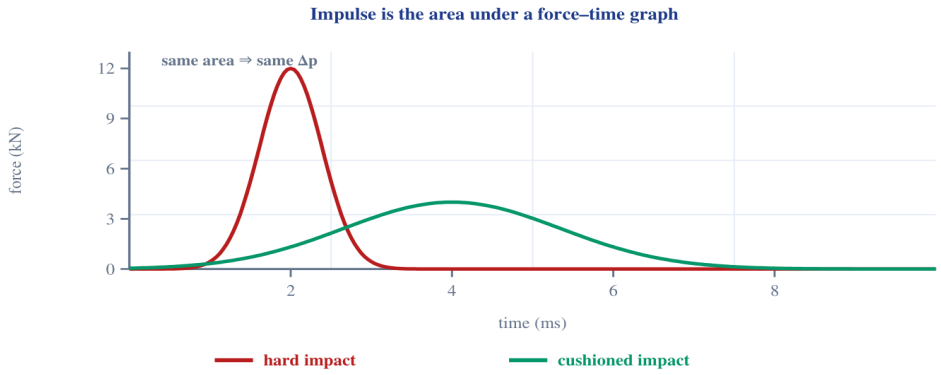
Figure. Ignoring friction, ½mv² = mg(h■ – h), so the speed peaks in the valleys and vanishes at the highest crest.

3. Momentum, impulse and collisions

Momentum is p = m v, and the impulse–momentum theorem says

$$\int F,dt = \Delta p. \tag{5}$$

The integral is the area under a **force–time** graph. This is why airbags work: the same Δ p spread over a longer time means a smaller peak force.



A short, large force and a long, small force can deliver the same impulse — but only one of them breaks things.

Figure. A short, large force and a long, small force can deliver the same impulse — but only one of them breaks things.

In a collision between two bodies with no external impulse, total momentum is conserved:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2. \tag{6}$$

In one dimension this is one equation with two unknowns, so you need a second piece of physics. The usual choice is the coefficient of restitution,

$$e = (v_2 - v_1)/(u_1 - u_2), \tag{7}$$

with e=1 elastic and e=0 perfectly inelastic. Solving the pair gives

$$v_1 = ((m_1 - e\ m_2)u_1 + (1+e)m_2u_2)/(m_1+m_2),\ v_2 = ((m_2 - e\ m_1)u_2 + (1+e)m_1u_1)/(m_1+m_2). \tag{8}$$

The kinetic energy lost is

$$\Delta K = (1)/(2),(m_1m_2)/(m_1+m_2)(1-e^2)(u_1-u_2)^2. \tag{9}$$

Everything about collisions is visible in that formula: the loss is largest for e = 0, vanishes for e = 1, and is proportional to the square of the **relative** speed.

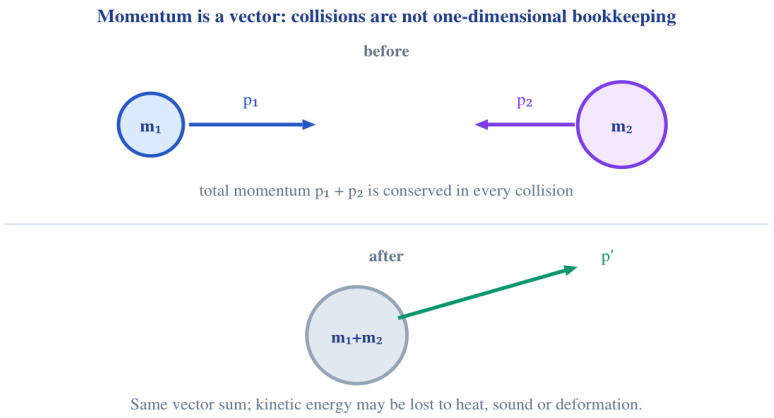


Figure 2. Momentum is conserved in every collision; kinetic energy is conserved only in elastic ones. Energy lost becomes heat, sound or deformation.

Interactive experiment: One-dimensional collisions, measured not guessed — open the online handout to use it.

4. Rotation: the same laws, one level up

For a rigid body about a fixed axis,

$$\tau = I\alpha,\ L = I\omega,\ K_{\{rot\}} = (1)/(2)\ I\omega^2, \tag{10}$$

with

$$I = \Sigma\ m_i\ r_i^2\ \{or\}\ I = \int r^2\ dm. \tag{11}$$

The parallel-axis theorem, $I = I_{(cm)} + Md^2$, lets you shift the axis once you know the centre-of-mass value.

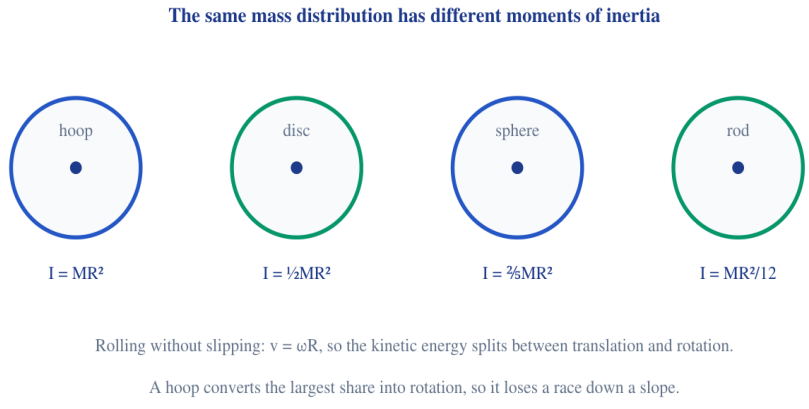


Figure 3. I is the rotational refusal to accelerate: it is to rotation what mass is to translation, and it depends on how far the mass sits from the axis.

Rolling without slipping ties the two motions together, $v = \omega R$, so

$$K = (1)/(2)\ mv^2 + (1)/(2)\ I\omega^2 = (1)/(2)\ mv^2(1 + (I)/(mR^2)). \tag{12}$$

That bracket is why a hoop, a disc and a sphere released together down a slope arrive in that order — and why the answer does not depend on the mass or the radius, only on the **shape**

factor $I/(mR^2)$.

5. Frames, fictitious forces, and when to leave the inertial frame

In a frame accelerating with a_0 relative to an inertial one, Newton's law needs an extra term:

$$m\,a' = F - m\,a_0. \tag{13}$$

The extra -m a_0 is a **fictitious force**. It is not an interaction; it is the price of doing bookkeeping in a non-inertial frame. The same logic produces the centrifugal force $-m\omega\times(\omega\times r)$ and the Coriolis force $-2m\omega\times v'$ in a rotating frame.

Use fictitious forces when the frame buys you simplicity — a bead in a groove on a turntable, a pendulum in an accelerating lift — and be explicit that they are not interactions.

Choosing the tool. Ask three questions in order. (1) Are the forces known at every instant? If yes, use Newton and acceleration. (2) Does the question only involve speeds and positions? Then use energy. (3) Are the forces brief, internal or unknown? Then use momentum. Write the choice down; examiners mark reasoning, not just answers.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Work by a varying force. A force $F(x) = 6x^2$ newtons acts on a 2.0 {kg} body initially at rest at x=0, where x is in metres. (a) Find the work done from x=0 to x=3.0 {m}. (b) Find the speed at x=3.0 {m}. (c) Find the power delivered at that instant.

Worked solution →

[1] Problem 2 — A block on a rough slope with a spring. A block of mass m slides down a slope of angle θ and compresses a spring of stiffness k by distance d before stopping. The coefficient of kinetic friction is μ. (a) Use energy to relate the distance slid and the compression. (b) Find the maximum compression if the block starts s metres up the slope from the uncompressed spring. (c) Show that the answer reduces correctly for μ=0 and check the dimensions.

Worked solution →

[2] Problem 3 — Elastic and inelastic collisions from one formula. A 2.0 {kg} trolley moving at 3.0 {m,s^(-1)} hits a stationary 1.0 {kg} trolley. (a) Find both final velocities for e=1. (b) Find them for e=0. (c) Find the kinetic energy lost for e=0 and for e=0.5, and verify the general formula $\Delta K = (1)/(2)\mu(1-e^2)(u_1-u_2)^2$ with the reduced mass $\mu = m_1m_2/(m_1+m_2)$.

Worked solution →

[2] Problem 4 — The ballistic pendulum. A bullet of mass m is fired into a block of mass M hanging on a string of length and embeds itself. The block swings up to height h. (a) Explain why momentum is conserved in the impact but mechanical energy is not. (b) Find the bullet's speed in terms of h. (c) Show that the fraction of kinetic energy lost is M/(m+M), and evaluate it for m = 10 {g}, M = 2.0 {kg}.

Worked solution →

[2] Problem 5 — Rolling race. A hoop, a solid disc and a solid sphere, all of mass M and radius R, are released from rest at the same height on a slope of angle θ and roll without slipping. (a) Find the acceleration of each. (b) Find the order of arrival and the ratio of the times. (c) Explain why mass and radius drop out, and what would change if the bodies slid frictionlessly instead.

Worked solution →

[2] Problem 6 — Two blocks and a pulley. Masses $m_1 = 3.0$ {kg} and $m_2 = 5.0$ {kg} hang from a light inextensible string over a light frictionless pulley. (a) Find the acceleration and the tension. (b) Check the limits $m_2 \to m_1$ and $m_1 \to 0$. (c) Find the tension if the pulley now has moment of inertia I and radius R, and show how the answer reduces when $I \to 0$.

Worked solution →

[2] Problem 7 — Explosion in mid-air. A shell of mass 3m moving horizontally at speed v explodes into two fragments of masses m and 2m. The lighter fragment is brought to rest by the explosion. (a) Find the velocity of the heavier fragment. (b) Find the energy released. (c) Where is the centre of mass four seconds later, and how does this illustrate momentum conservation?

Worked solution →

[2] Problem 8 — The falling chain. A uniform chain of mass M and length L hangs with its lower end just touching a scale and is released. (a) Show that when a length x has landed the reading is 3Mgx/L at the instant of impact of a new link, and explain the factor 3. (b) Find the reading once the whole chain rests. (c) Discuss briefly why the reading is **not** simply the weight of the landed part.

Worked solution →

[2] Problem 9 — Angular momentum and the skater. A skater with moment of inertia I_1 spins at angular speed ω_1 with arms out. She pulls her arms in, reducing her moment of inertia to $I_2 = 0.4I_1$. (a) Find her new angular speed and explain why angular momentum is conserved. (b) Find the work done by her muscles. (c) Explain where that energy comes from and why the process cannot run backwards spontaneously.

Worked solution →

[3] Problem 10 — A variable-mass rocket. A rocket of initial mass m_0 burns fuel at rate $m = -\alpha$ and ejects it at speed u relative to the rocket. (a) Derive the rocket equation by applying momentum conservation over a short interval, and show that $\Delta v = u \ln(m_0/m_f)$ in the absence of gravity. (b) Include gravity and show that the speed gained is reduced by gt. (c) Estimate the mass ratio needed for a rocket to reach orbital speed 7.8 {km,s^(-1)} with u = 3.0 {km,s^(-1)}, and comment on the practical consequence.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

HyperPhysics mechanics map

A compact concept map for revising definitions and standard results quickly.